HandsMen Threads: Elevating the Art of Sophistication in Men's Fashion

User story :

HandsMen Threads, a dynamic organization in the men's fashion industry, is implementing a Salesforce-based platform to modernize data management, streamline business operations, and elevate customer relations. As a Sales Representative, I want to track customer purchases, preference profiles, and order status in one central application, so that I can provide personalized styling recommendations and deliver a seamless sales experience. As a Warehouse / Inventory Manager, I want automated alerts whenever product stock levels drop below thresholds, so that I can initiate timely restocking and prevent stockout delays. As a Customer Experience Manager, I want post-order email updates and dynamic loyalty program status updates, so that customers stay engaged and rewarded for repeat business. As a Finance / Operations Administrator, I want scheduled nightly bulk updates to capture large order adjustments and update financial records automatically, so that daily financial and inventory reports are always accurate and ready for management decision-making.

1. INTRODUCTION

1.1 Project Overview :

HandsMen Threads is embarking on a Salesforce platform project designed to revolutionize its data management and enhance customer relations. The system brings together customer profiles, order history, inventory, sales transactions, and loyalty tracking into a single custom Salesforce application.

By leveraging core Salesforce features—including Custom Objects, Page Layouts, Validation Rules, Record-Triggered Flows, Email Templates, Apex Triggers, and Asynchronous/Batch Apex—HandsMen Threads ensures high data integrity directly from the User Interface (UI). The platform automates transactional tasks such as sending immediate order confirmations, evaluating dynamic loyalty tiers, alerting warehouse staff of low stock, and executing scheduled midnight batch jobs for bulk order processing.

1.2 Purpose:

The primary purpose of the HandsMen Threads Salesforce project is to eliminate fragmented data and manual processes across sales, inventory, and customer outreach. The system ensures:

- **Centralized Fashion & Customer Data:** Single repository for customer preferences, purchase histories, product inventories, and sales orders.
- **Streamlined Fulfillment Workflows:** Every customer order follows an automated pipeline from confirmation to inventory deduction and shipping setup.
- **Automated Data Quality & Validation:** Validation Rules and UI enforcement safeguard accuracy and consistency across all business records.

- **Enhanced Customer Engagement & Operational Alerts:** Automated email triggers keep customers updated while proactive stock alerts prevent operational bottlenecks.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 Sales Representative	Deliver personalized fashion recommendations and manage orders	Customer histories and order details are spread across separate legacy files	Systems are unintegrated and require manual updates	Frustrated and slow to respond to high-value clients
PS-2 Warehouse Manager	Maintain adequate product stock for popular apparel lines	Stockouts occur unexpectedly and inventory tracking is reactive	Inventory changes are checked manually at the end of the day	Anxious about delayed shipments and missed sales
PS-3 Customer Experience Manager	Increase customer retention and reward loyal shoppers	Order confirmations and loyalty tier updates are delayed or missed	Customer tier assignment requires manual purchase aggregation	Concerned about customer churn and poor brand perception
PS-4 Administrator / Operations Lead	Keep inventory counts and financial records synchronized daily	Nightly reconciliation takes hours and causes start-of-day discrepancies	Bulk orders must be recalculated manually outside of business hours	Overwhelmed and uncertain about data reliability

2.2 Empathy Map Canvas

User Persona: Customer Experience & Sales Representative

- **Think & Feel:** Anxious about overselling out-of-stock garments. Eager to provide top-tier, personalized service to premium buyers.
- **Hear:** Feedback from customers asking for immediate shipping status; inquiries from management about daily sales totals.
- **See:** Multiple spreadsheets, legacy sales receipts, and manual inventory tracking logs with no single view of client status.
- **Say & Do:** Manually checks inventory registers before placing orders, emails order receipts individually, and manually calculates total spend for loyalty eligibility.

2.3 Brainstorming & Idea Prioritization

Step 1: Brainstormed Ideas

- **Spreadsheet Trackers:** Shared cloud spreadsheets for inventory, customer spend, and order history.
- **Third-Party Standalone Portal:** A custom e-commerce portal integrated loosely via external webhooks.

- **Salesforce Platform (HandsMen Threads App):** Centralized cloud app using Custom Objects, Flows, Apex, Validation Rules, Reports, and Batch Jobs.
- **Automated Customer Outreach:** Email templates triggered immediately upon order confirmation.
- **Scheduled Batch Apex Jobs:** Nightly processing for bulk order financial updates and inventory adjustments.

Step 2: Grouping

Group	Ideas	Description
Manual solutions	Spreadsheet Trackers	Quick to implement, but vulnerable to human error and duplicate records.
Semi-automated solutions	Third-Party Standalone Portal	Automates online bookings, but creates data silos requiring complex external integration.
Fully automated cloud solutions	Salesforce Platform App, Automated Outreach, Scheduled Batch Jobs	End-to-end cloud platform covering data modeling, business automation, UI validation, and scalable batch processing.

Step 3: Idea Prioritization

Each idea is rated High, Medium, or Low against the criteria below.

Idea	Impact	Feasibility	Cost Efficiency
Spreadsheet Trackers	Low	High	High
Third-Party Standalone Portal	Medium	Medium	Low
Salesforce Platform App	High	High	High
Automated Customer Outreach	High	High	High
Proactive Low-Stock Alerts	High	High	High
Scheduled Batch Apex Updates	High	Medium	High

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stage 1: Product Browsing & Order Creation

- **User Actions:** Customer selects items; Sales Rep enters order details into Salesforce.
- **Touchpoints:** HandsMen Threads Salesforce Lightning App, Sales Rep.
- **Pain Points:** Incorrect pricing or missing customer details.
- **Solution:** Custom Sales_Order__c object with required fields and Validation Rules.
- **Emotion:** Enthusiastic

Stage 2: Order Confirmation & Notification

- **User Actions:** Order status changes to "Confirmed".

- **Touchpoints:** Automated Email Gateway, Customer Inbox.
- **Pain Points:** Delayed receipt causing customer anxiety.
- **Solution:** Record-Triggered Flow sends an instant automated order confirmation email.
- **Emotion:** Reassured.

Stage 3: Dynamic Loyalty Processing

- **User Actions:** System recalculates customer spend based on order confirmation.
- Touchpoints: Customer_Profile__c record, Loyalty Engine.
- **Pain Points:** Manual calculation errors and delayed perk distribution.
- **Solution:** Record-Triggered Flow / Apex Trigger updates customer loyalty status dynamically.
- **Emotion:** Appreciative.

Stage 4: Proactive Stock Monitoring

- **User Actions:** Order decreases inventory count on product records.
- **Touchpoints:** Inventory Database, Warehouse Email.
- **Pain Points:** Inventory dropping to zero unexpectedly.
- **Solution:** Record-Triggered Flow checks if quantity drops below 5 units and immediately emails the warehouse team.
- **Emotion:** Confident

Stage 5: Scheduled Midnight Processing

- **User Actions:** System executes daily midnight batch jobs for bulk orders.
- **Touchpoints:** Asynchronous Apex, Financial Ledger Records.
- **Pain Points:** System slowdown during business hours when processing high volumes.
- **Solution:** Scheduled Batch Apex processes bulk updates at midnight to maintain system performance.
- **Emotion:** Satisfied.

3.2 Solution Requirements

Functional Requirements

- **FR-1: Data Modeling & UI:** Custom objects for Customers, Products, Orders, and Order Items with Lightning App pages and custom tabs.
- **FR-2: Automated Order Confirmation:** Instant email notifications sent to customers upon order submission.
- **FR-3: Dynamic Loyalty Program:** Automated recalculation and update of customer loyalty tiers based on lifetime purchase history.
- **FR-4: Proactive Stock Alerts:** Automatic notification emails sent to warehouse staff when product stock drops below 5 units.

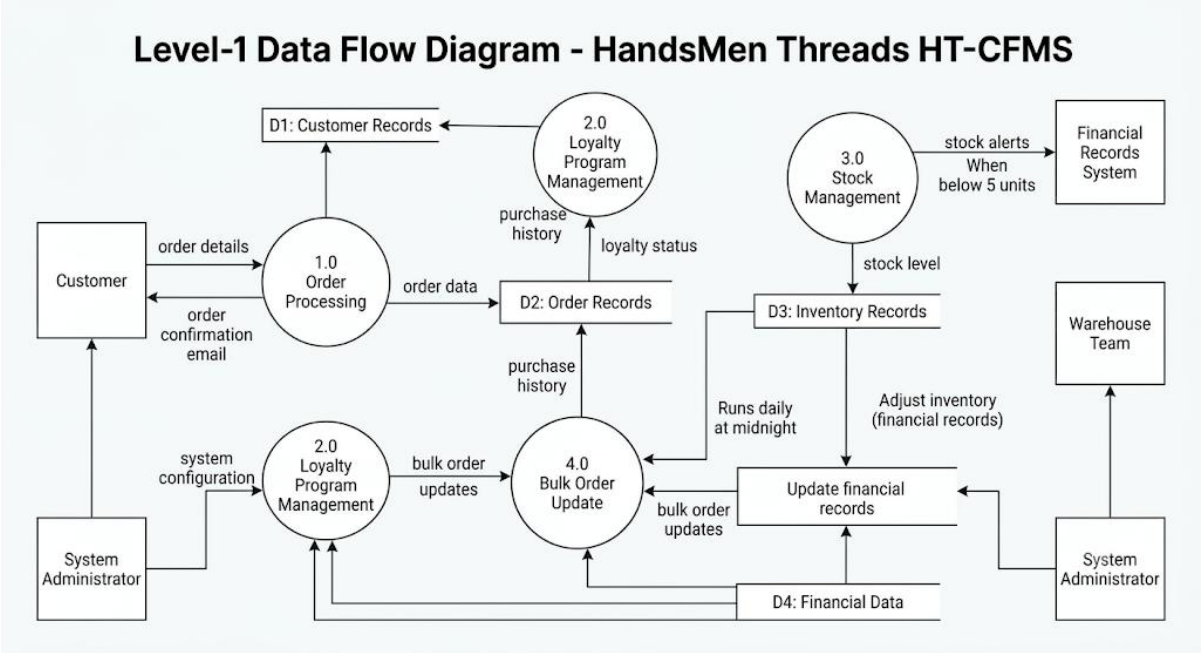
- **FR-5: Scheduled Bulk Order Updates:** Asynchronous Batch Apex running daily at midnight to update financial ledgers and balance stock.

Non-Functional Requirements

- **NFR-1 Security:** Profile and Role-based security ensuring appropriate access levels across Sales, Warehouse, and Management teams.
- **NFR-2 Usability:** Clean, intuitve Lightning App layouts requiring minimal staff training.
- **NFR-3 Performance:** Asynchronous execution of batch tasks to avoid UI lag and governor limit issues during operational hours.
- **NFR-4 Scalability:** Built on Salesforce cloud architecture to handle high transaction volumes effortlessly.
- **NFR-5 Data Integrity:** Point-of-entry Validation Rules and Apex Triggers to prevent invalid data input.

3.3 Data Flow Diagram

This is the Level-1 data flow diagram for the main steps of the system.



Here is how the diagram maps to the steps:

- **Step 1 (Input & Storage):** Sales Rep enters order details. Validation rules check fields before saving to Customer__c, Sales_Order__c, and Product__c objects.
- **Step 2 (Automation Execution):** Upon order confirmation, the Record-Triggered Flow triggers parallel processing paths.

- **Step 3 (Customer Engagement):** System retrieves customer contact info and sends an Order Confirmation Email.
- **Step 4 (Loyalty Calculation):** System aggregates purchase totals and updates the customer's loyalty tier.
- **Step 5 (Inventory Monitoring):** System checks remaining stock. If quantity falls below 5, an automated email alert goes out to warehouse staff.
- **Step 6 (Scheduled Processing):** At midnight, Batch Apex processes queued bulk order updates and synchronizes financial totals.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem–Solution Mapping

Problem (User Behavior / Constraint)	HandsMen Threads Solution	Salesforce-Native Capability
Scattered customer & order data	Centralized fashion data model	Custom Objects (Customer_Profile__c, Product__c, Sales_Order__c)
Data entry errors & invalid formats	UI point-of-entry validation	Validation Rules & Field Constraints
Missing post-order communications	Automated order receipts	Email Templates & Record-Triggered Flows
Delayed loyalty status assignment	Dynamic calculation based on spend	Record-Triggered Flows / Apex Triggers
Unannounced stockouts	Automated warehouse stock alerts	Record-Triggered Flows (Stock Threshold < 5)
Unauthorized access to sensitive records	Users see only what their role permits	Profiles, permission sets, roles, and sharing rules
System lag during bulk order processing	Off-peak scheduled updates	Asynchronous & Scheduled Batch Apex
Unrestricted access to sensitive data	Role-based data control	Roles, Profiles, Permission Sets, Sharing Rules

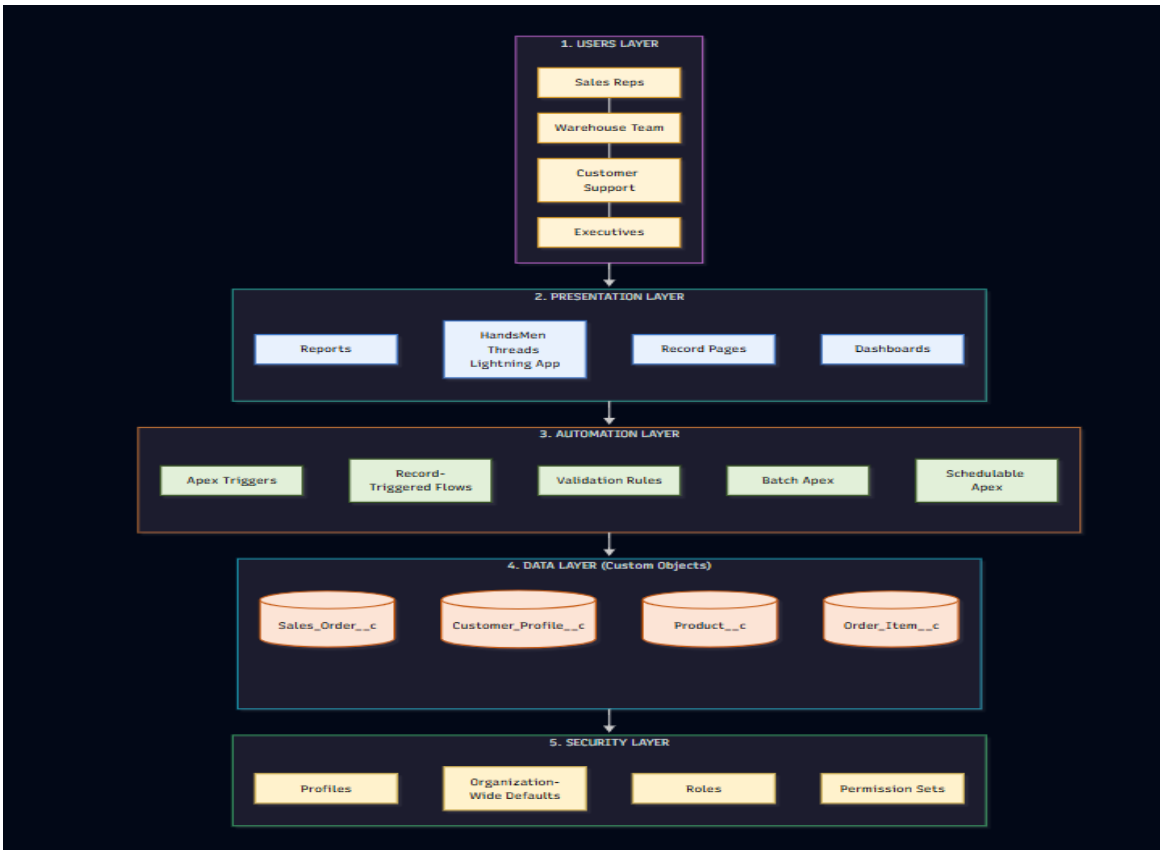
4.2 Proposed Solution

Parameter	Description
Problem	Disjointed sales tracking, manual inventory checks, delayed customer communication, and potential system slowdown during bulk processing.
Solution	A custom Salesforce application unifying data management, UI validation, automated workflows, and scheduled asynchronous processing.
Novelty	End-to-end integration of fashion order management with proactive inventory threshold alerts and automated nightly batch operations on one platform.
Social Impact	Professionalized operations, reduced material waste from stock inaccuracies, and higher workplace productivity.

Parameter	Description
Revenue Model	Enterprise apparel management framework with reduced operational overhead and improved client retention.
Scalability	Cloud-native platform capable of onboarding multi-store retail branches without core architectural re-engineering.

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers. Users interact through the Lightning interface, automation processes their requests, and the results are stored in custom objects. A security layer controls access at every level.



Here is how the layers work together:

- **Users Layer:** Persona-based interaction point tailored for Sales Reps, Warehouse Staff, Support Agents, and Business Managers.
- **Presentation Layer:** Lightning App navigation, customized Page Layouts, custom object tabs, and real-time analytical Dashboards.
- **Automation Layer:** Declarative Record-Triggered Flows for email alerts, Apex Triggers for complex checks, and Batch Apex for midnight execution.

- **Data Layer:** Relational schema comprising Custom Objects, Master-Detail links, Lookup relationships, and Roll-up Summary fields.
- **Security Layer:** Multi-tiered defense using Organization-Wide Defaults (OWD), Role Hierarchy, Profiles, and Permission Sets.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project follows the **Agile methodology**. Work is organized as **Epics → User Stories → Story Points** and delivered in **sprints**. Velocity from completed sprints is used to estimate the remaining work.

Product Backlog and Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	Set up developer edition org and developer environment.	1	High	Member1
Sprint-1	Data Modeling	USN-2	Create custom objects (Customer_Profile, Product, Sales_Order, Order_Item).	3	High	Member2
Sprint-2	Data Modeling	USN-3	Create custom tabs and navigation styles for all objects.	3	High	Member3
Sprint-2	Data Modeling	USN-4	Build custom fields (Picklists, Formulas, Currency, Lookup/Master-Detail).	5	High	Member4
Sprint-2	App Building	USN-5	Configure HandsMen Threads Lightning App and assign profile access.	3	High	Member4
Sprint-2	Data Modeling	USN-6	Define relationships (Master-Detail for Order-Items, Lookups for Orders)	5	High	Member1
Sprint-2	Page Layouts	USN-7	Design tailored page layouts for Sales and Warehouse teams.	5	High	Member2
Sprint-3	Automation	USN-8	Implement Validation Rules for UI data accuracy.	3	High	Member4
Sprint-3	Automation	USN-9	Create email templates and Record-Triggered Flow for order confirmations.	3	High	Member4
Sprint-3	Automation	USN-10	Build Record-Triggered Flow for dynamic loyalty program status updates.	5	High	Member1
Sprint-4	Automation	USN-11	Build Record-Triggered Flow for stock alerts when quantity drops below 5.	3	High	Member2
Sprint-5	Development	USN-12	Develop Apex Triggers for complex data verification.	5	High	Member3
Sprint-5	Development	USN-13	Develop Asynchronous / Batch Apex for scheduled bulk order updates at midnight.	5	High	Member4
Sprint-5	Security	USN-14	Configure roles, profiles, and permission sets for security setup.	5	High	Member 1
Sprint-6	Analytics	USN-15	Create custom reports and operational dashboards.	4	High	Member 2

Epic Summary

Epic	Stories	Story Points
Developer Setup	USN-1	1
Data Modeling & Design	USN-2 to USN-7	24
Automation	USN-8 to USN-11	14
Apex Development	USN-12 to USN-13	10
Security	USN-14	5
Analytics & Reports	USN-15	4
Total	**15 stories**	**58**

Sprint Schedule

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	4	7 days	5 Oct 2026	11 Oct 2026	4	11 Oct 2026
Sprint-2	21	7 days	12 Oct 2026	18 Oct 2026	21	18 Oct 2026
Sprint-3	11	7 days	19 Oct 2026	25 Oct 2026	11	25 Oct 2026
Sprint-4	8	7 days	26 Oct 2026	1 Nov 2026	8	1 Nov 2026
Sprint-5	10	7 days	2 Nov 2026	8 Nov 2026	10	8 Nov 2026
Sprint-6	4	7 days	9 Nov 2026	15 Nov 2026	4	15 Nov 2026
Total	**58**	**42 days**			**58**	

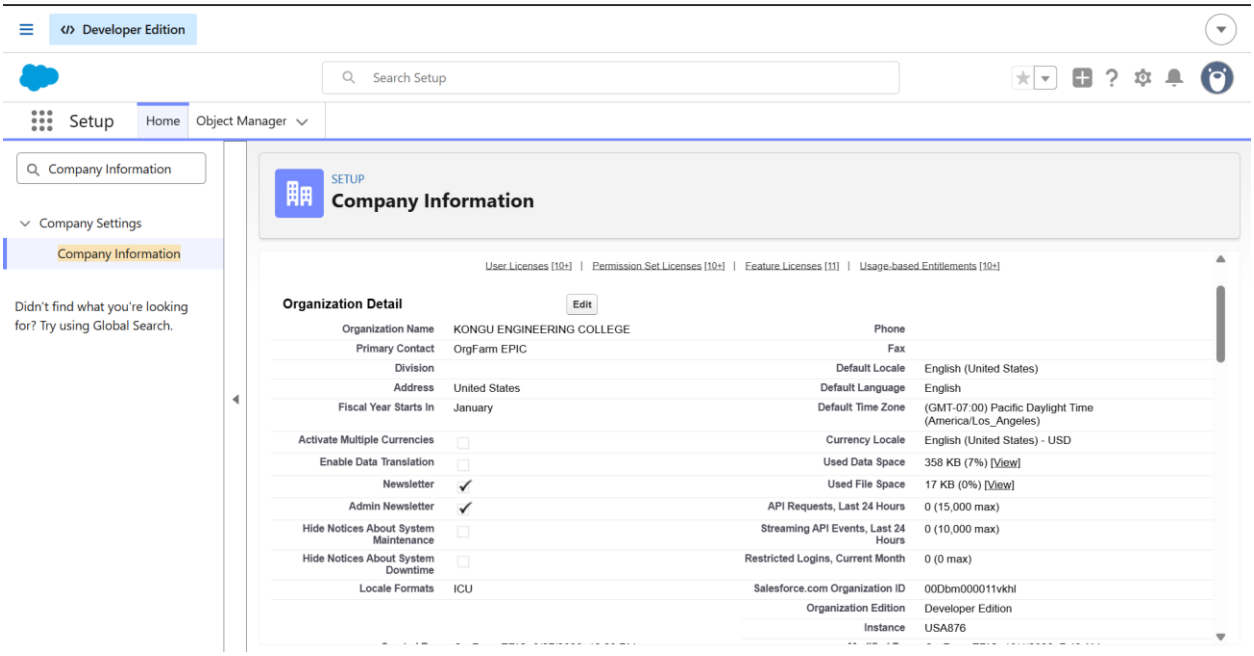
Team Workload

Member	Stories	Total Story Points
Member1	USN-1, USN-6, USN-10,USN-14	14
Member2	USN-2, USN-7, USN-11,USN-15	15
Member3	USN-3, USN-12	8
Member4	USN-4, USN-5, USN-8, USN-9, USN-13	21

6. Project Development Phase

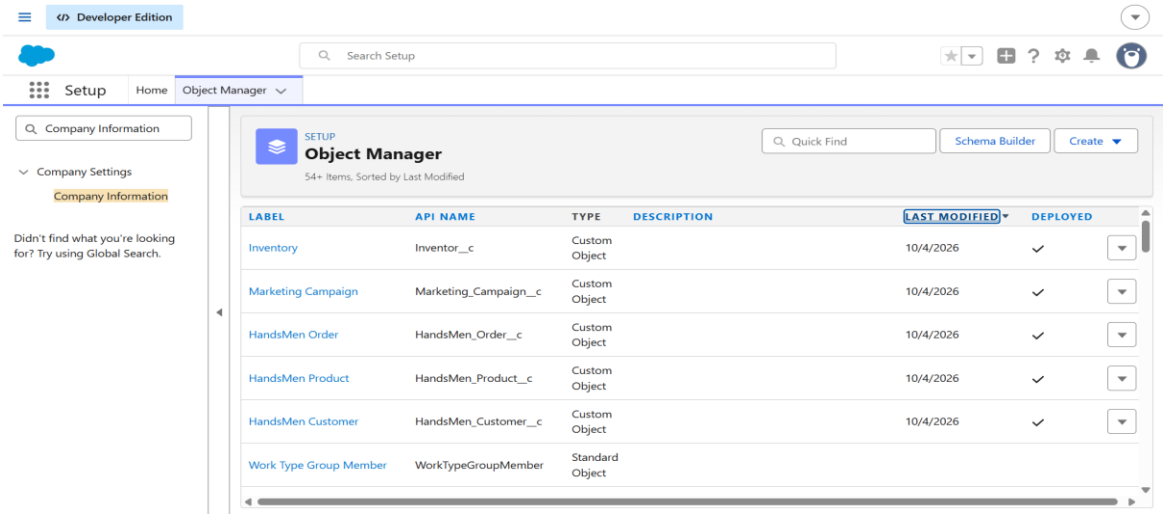
Milestone 1: Salesforce Developer Account Creation

- **Definition:** A free Salesforce Developer Edition org serves as the environment for building and testing the application.
- **Key Steps**: Sign up at developer.salesforce.com, verify email, log in, and configure administrative preferences.



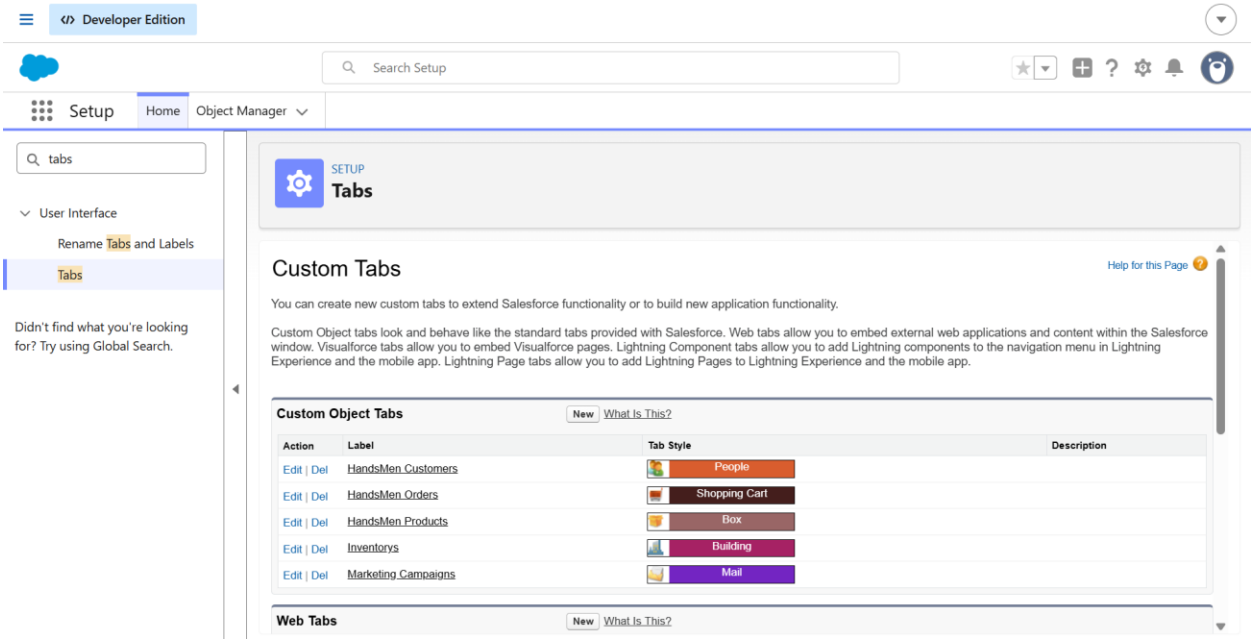
Milestone 2: Custom Object Creation

- **Definition**: Creating core custom tables: Customer_Profile__c, Product__c, Sales_Order__c, and Order_Item__c.
- **Key Steps:** Navigate to Object Manager, define labels, singular/plural names, record formats, and enable reporting options.



Milestone 3: Tabs Creation

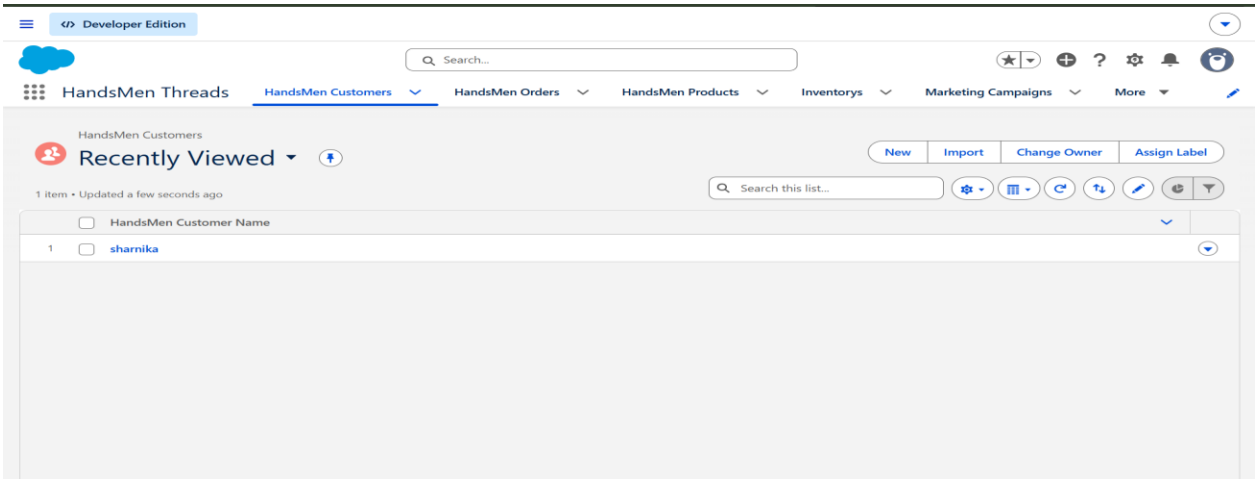
- **Definition:** Dedicated UI navigation tabs for quick access to core data objects.
- **Key Steps:** Create object tabs in Setup, assign distinct tab styles, and configure profile visibility settings.



Milestone 4: Lightning App Creation

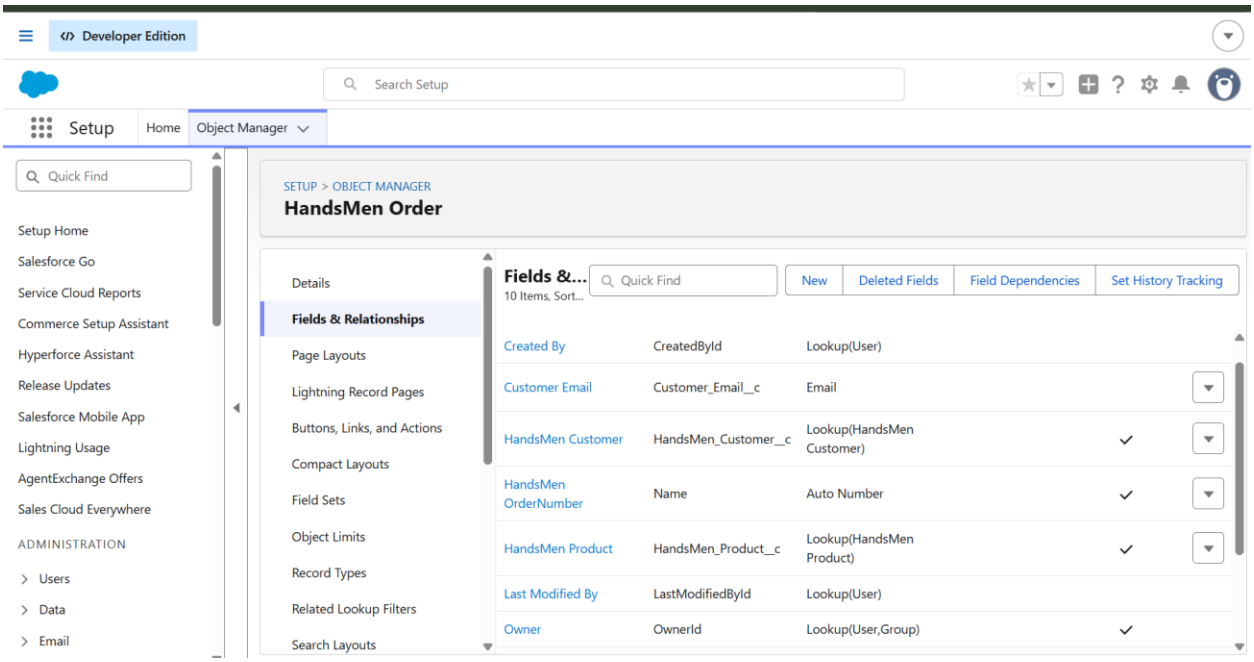
- **Definition:** A branded Lightning Application titled "HandsMen Threads" bundling object tabs, utility items, and analytics.

- **Key Steps:** Use App Manager to build a standard navigation Lightning App, brand with company styling, add custom tabs, and assign to user profiles.



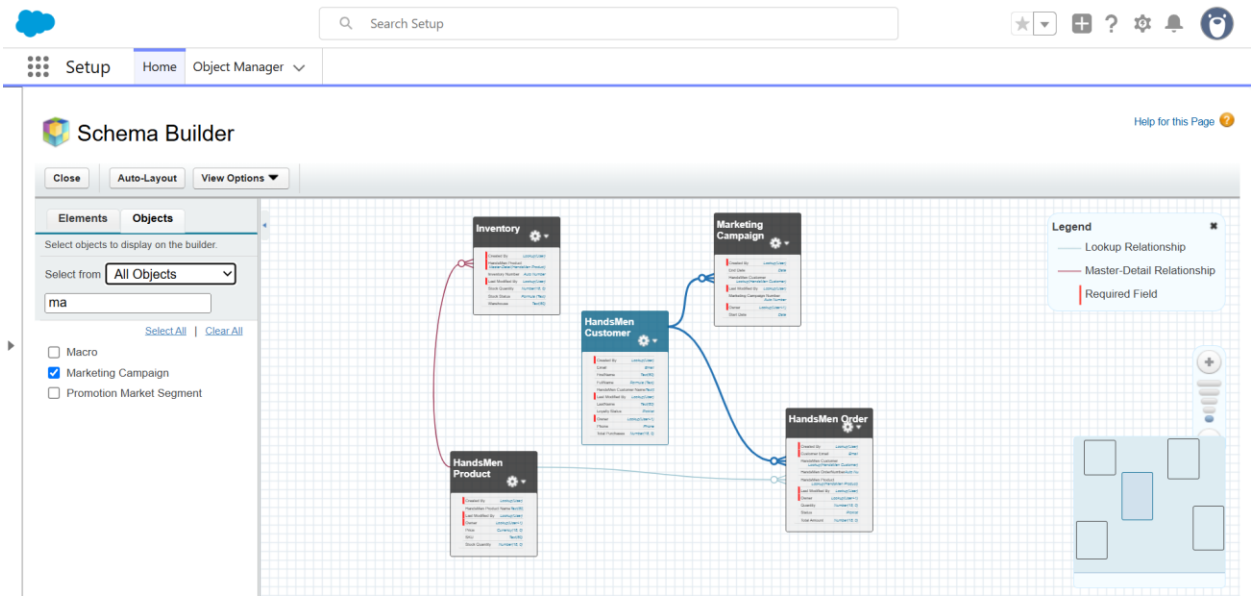
Milestone 5: Fields Creation

- **Definition:** Adding data attributes across objects to record inventory levels, prices, statuses, and customer categories.
- **Key Steps:** Add custom fields (Currency, Number, Text, Picklists, Formulas) with explicit UI field requirements.



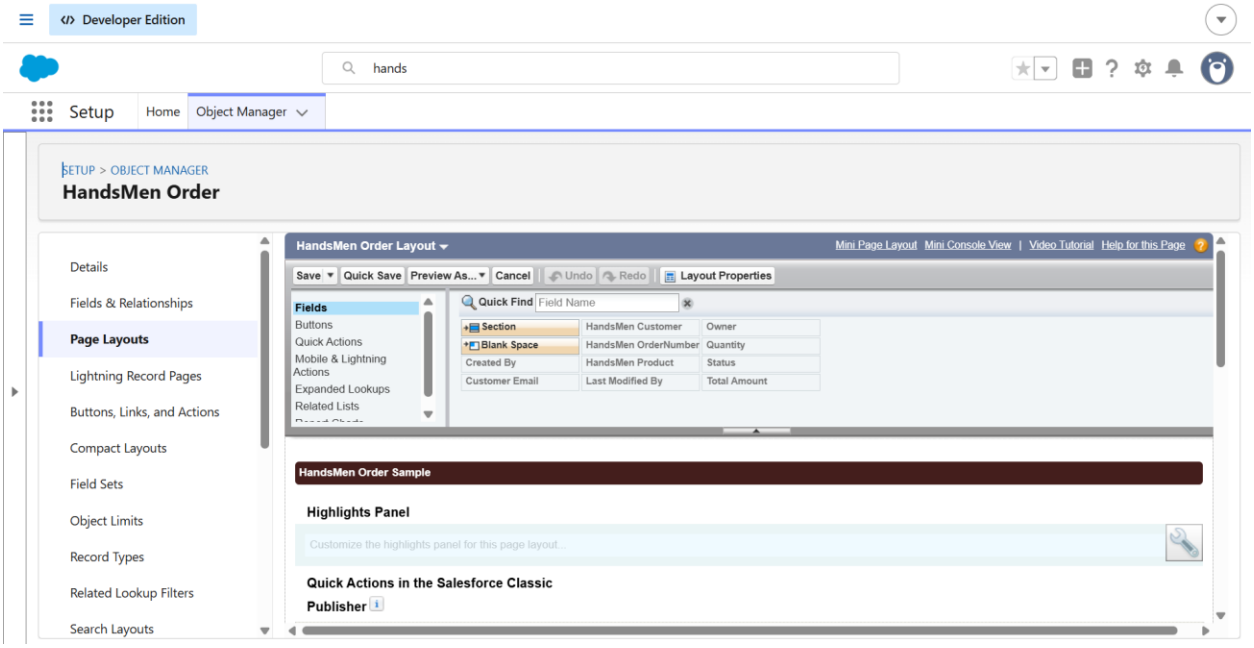
Milestone 6: Create Relationships

- **Definition:** Linking records across objects to structure order lineups and customer histories.
- **Key Steps**: Build Master-Detail relationships between Order_Item__c and Sales_Order__c, and Lookup relationships connecting orders to customers and products.



Milestone 7: Page Layout Customization

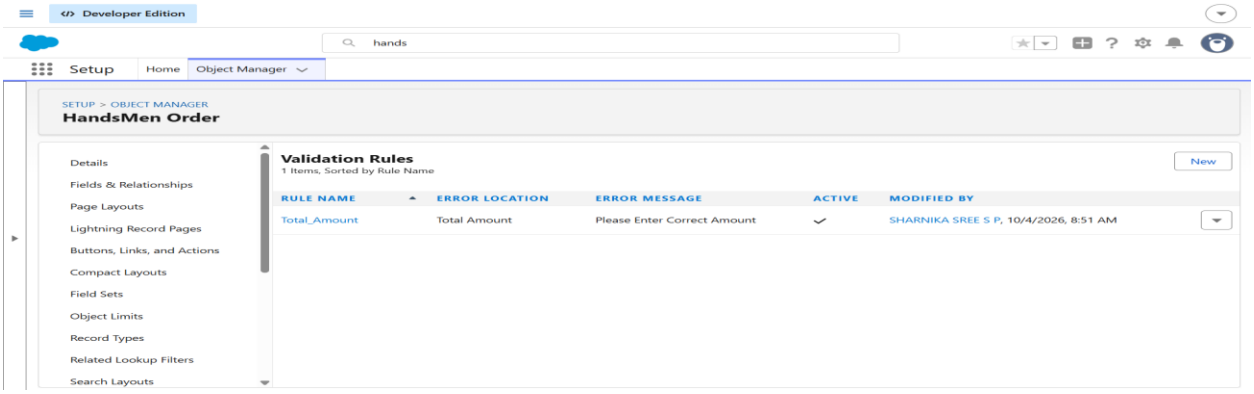
- **Definition:** Tailoring layout screens to display relevant fields clearly based on user roles.
- **Key Steps:** Group fields into organized sections, adjust field accessibility settings, and place related list views on primary record pages.

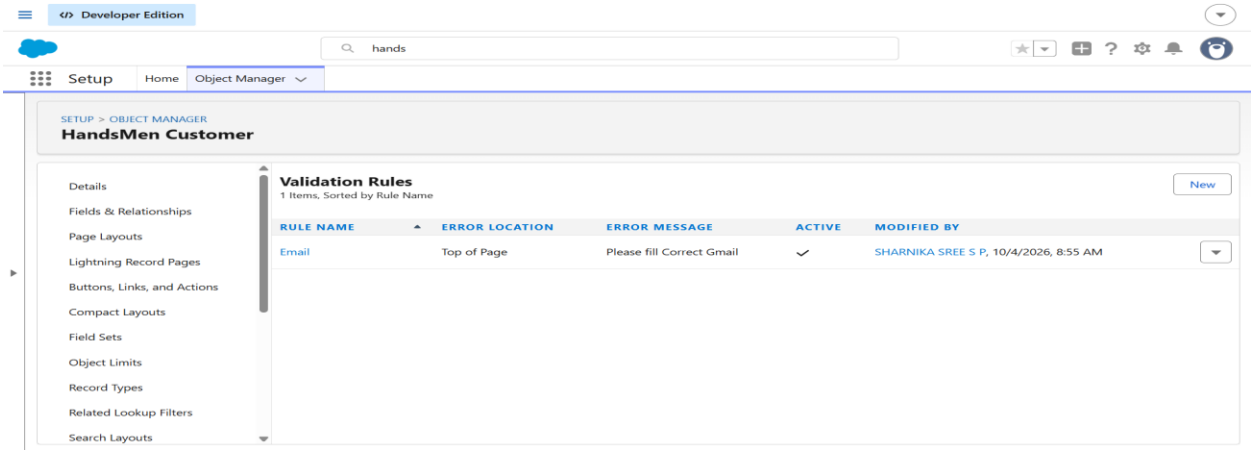


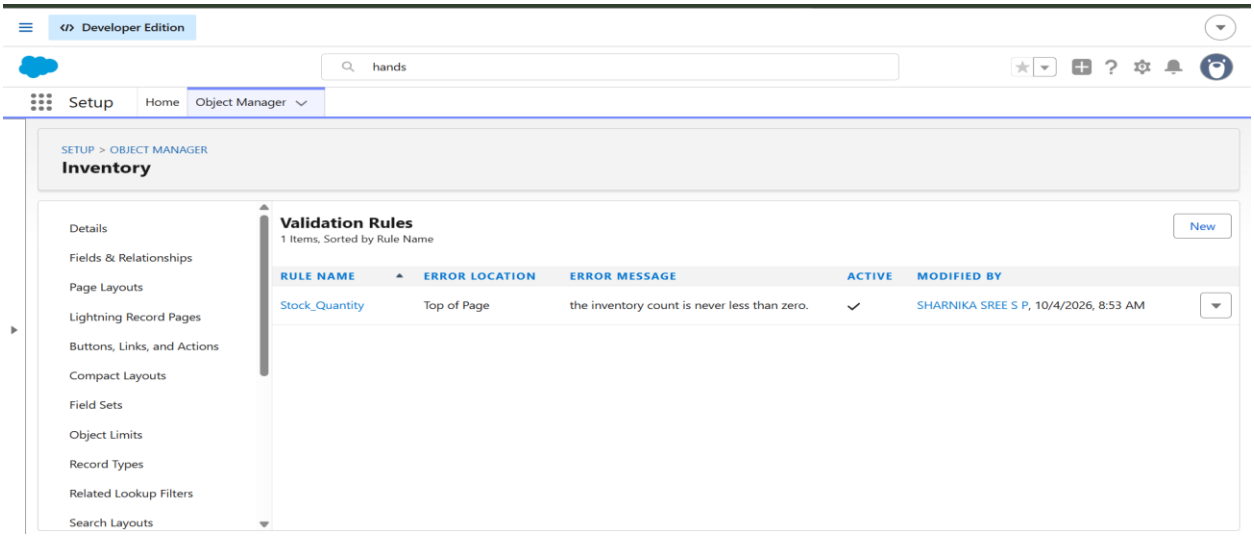
Milestone 8: Validation Rules

- **Definition:** Declarative formulas preventing illegal data entries directly at the UI level.

- **Key Steps:** Add validation logic for non-negative stock numbers, valid contact details, and required order dates. Activate and verify error messages.

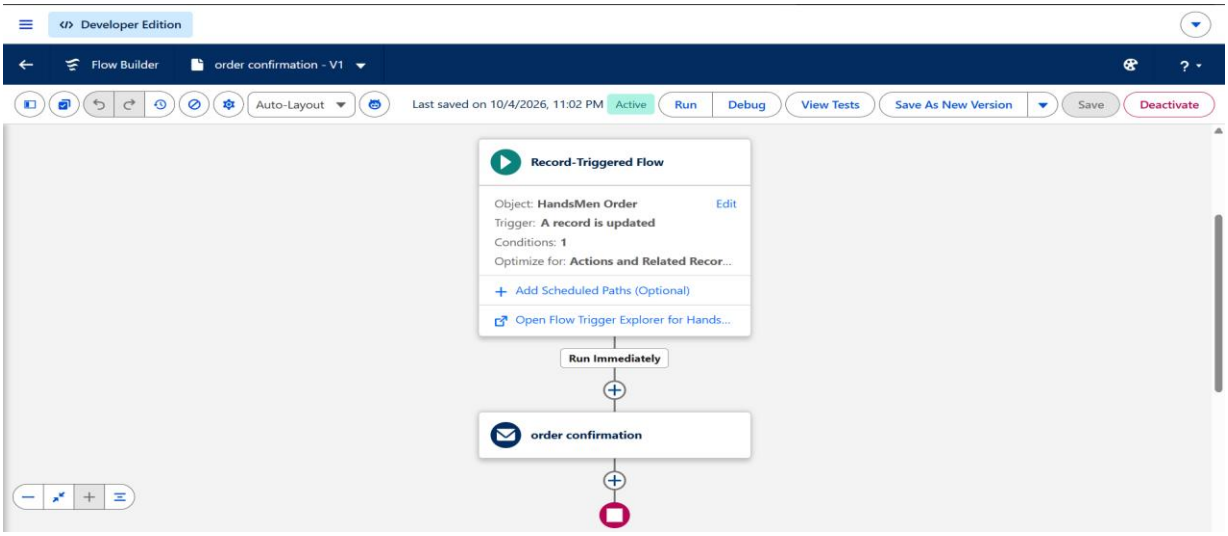






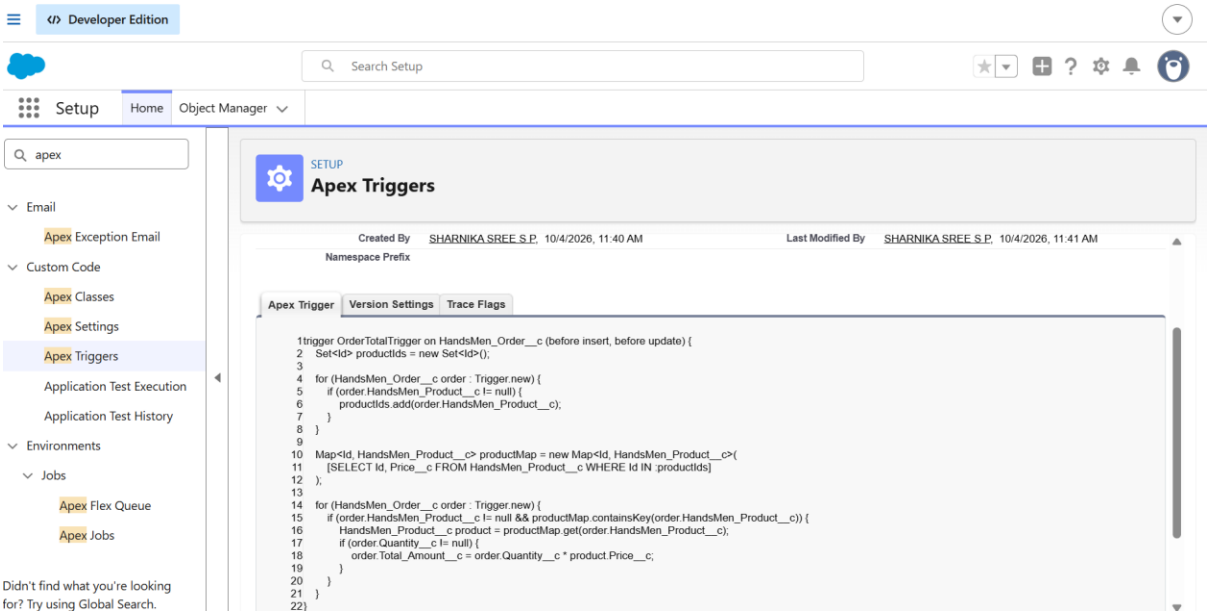
Milestone 9: Record-Triggered Flows & Email Templates

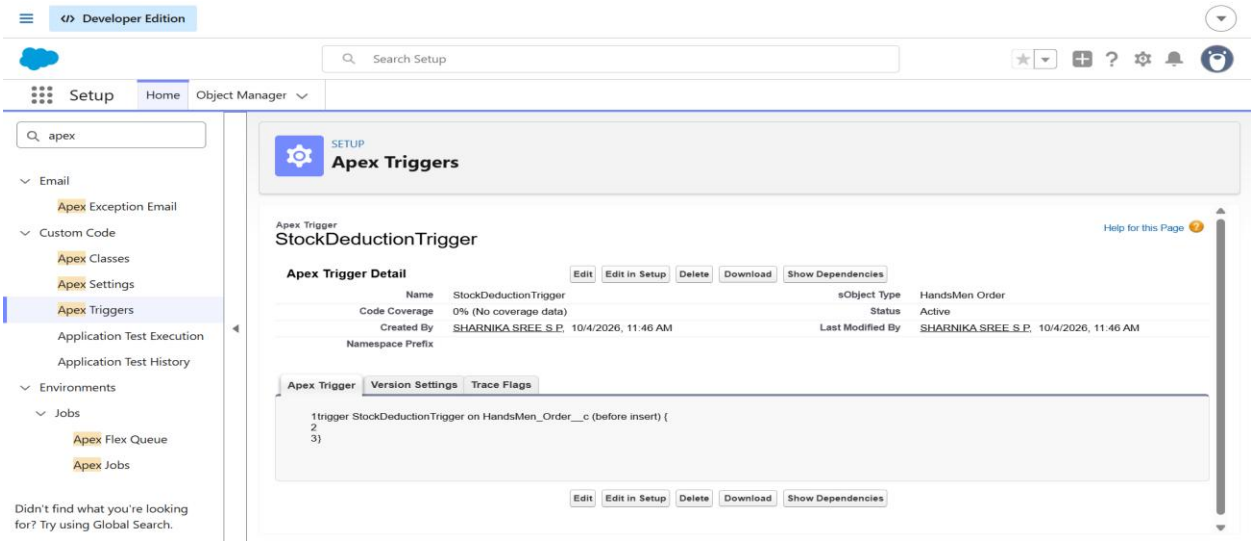
- **Definition**: Automating post-order emails, stock alerts, and loyalty tier updates without custom code.
- **Key Steps**: Build email templates. Configure Record-Triggered Flows on Sales_Order__c and Product__c for automated customer order emails, loyalty status updates, and low-stock alerts (< 5 units) to warehouse staff.



Milestone 10: Apex Triggers logic

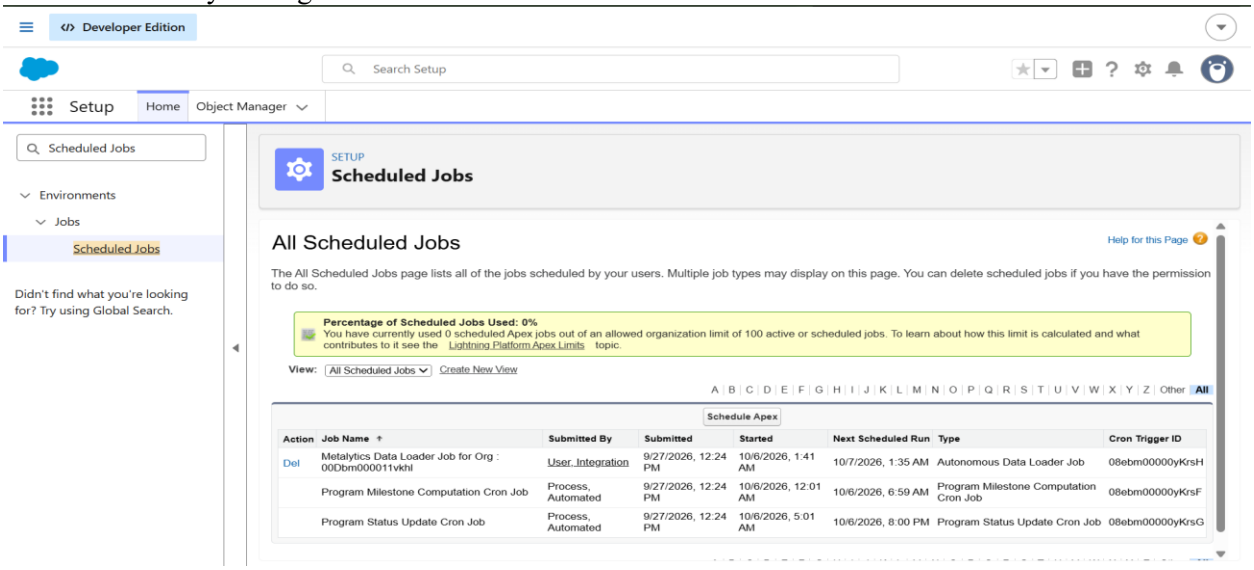
- **Definition:** Custom Apex logic for advanced business rules and validation constraints exceeding declarative capabilities.
- **Key Steps:** Implement Apex Triggers on order insertion/update events, ensuring governor-limit compliant bulk handling with test class execution.





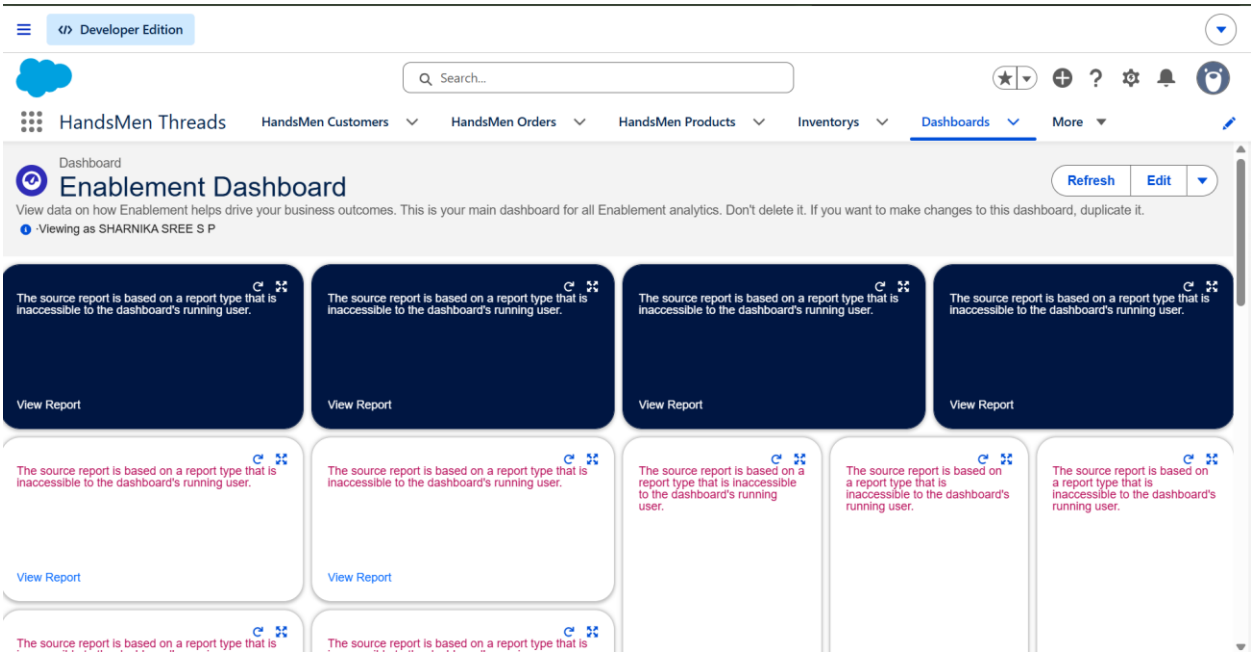
Milestone 11: Asynchronous & Scheduled Batch Apex

- **Definition:** Background processes running during off-peak hours to process bulk order updates and adjust ledgers without impacting UI performance.
- Key Steps: Write a Database.Batchable and Schedulable Apex class. Test via Developer Console and schedule daily midnight runs.



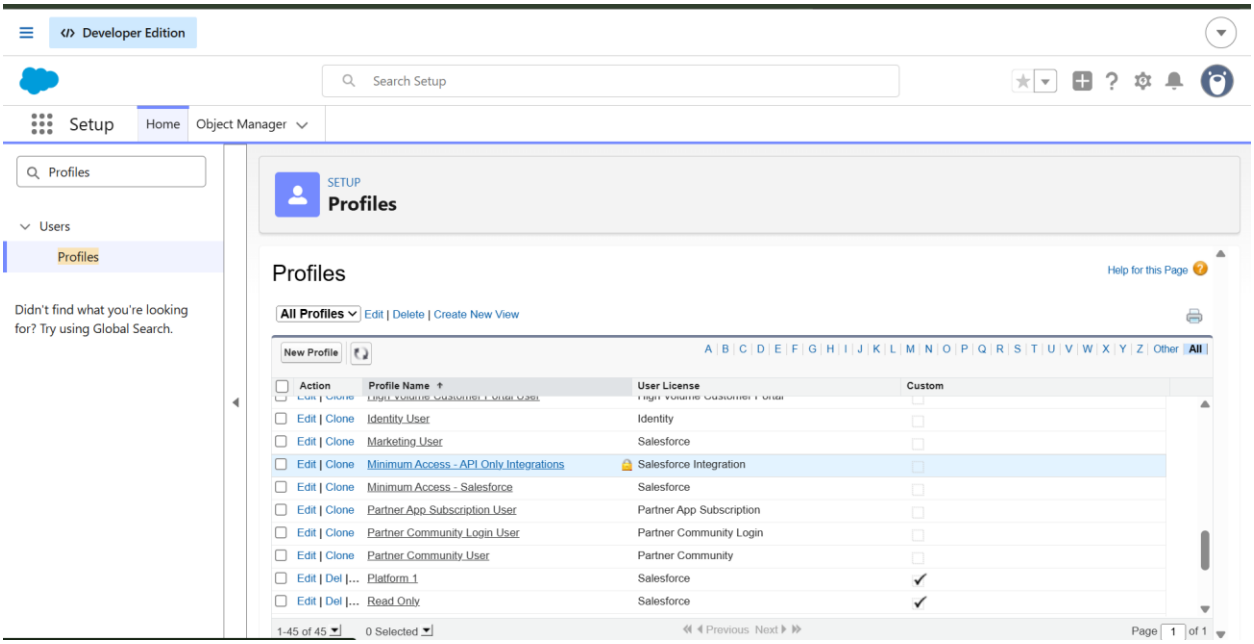
Milestone 12: Reports & Dashboards

- **Definition:** Real-time visibility into sales metrics, inventory counts, loyalty distributions, and order performance.
- **Key Steps:** Create custom report types, aggregate metrics by category, and build visual dashboards with dynamic chart components.



Milestone 13: Security Setup

- **Definition:** Safeguard business records across user groups using role hierarchies, profiles, and permission sets.
- **Key Steps:** Configure Organization-Wide Defaults (OWD), build Role Hierarchies, refine profile permissions for Sales and Warehouse teams, and test access via "Login As".



Milestone 14: Conclusion

Project Summary

The HandsMen Threads project establishes a unified, cloud-based platform on Salesforce, replacing fragmented manual records. By integrating data modeling, automated email communications, dynamic loyalty tracking, proactive low-stock notifications, and midnight batch jobs, the platform delivers a reliable, end-to-end operational workflow.

Outcomes and Benefits

- **Operational Efficiency:** Automated order notifications and stock alerts eliminate manual cross-checking.
- **High Data Accuracy:** Point-of-entry Validation Rules and Apex Triggers keep database records consistent.
- **Proactive Inventory Management:** Real-time notifications for stock levels under 5 units help prevent stockouts.
- **System Performance Optimization:** Nightly asynchronous Batch Apex jobs process bulk operations during off-peak hours without degrading user performance.
- **Scalability & Security:** Robust role hierarchies and declarative capabilities support business growth and secure data access.

Possible Future Enhancements

- **Customer Self-Service Portal:** Experience Cloud site allowing clients to track order statuses, view loyalty points, and update profiles directly.
- **Omnichannel Notifications:** SMS and messaging gateway integrations (e.g., WhatsApp Business API) for instant delivery updates.
- **E-Commerce Integration:** REST API connections to online web stores for seamless inventory synchronization.
- **AI-Driven Analytics:** Salesforce Einstein integration to forecast apparel demand trends and personalize customer styling recommendations.

Closing Statement

The HandsMen Threads Care & Fashion Management System establishes a secure, scalable, and configuration-first foundation. The solution empowers HandsMen Threads to scale retail operations seamlessly while delivering exceptional service to every client .